

Midterm 2 W26

2026-03-19

Instructions

This is an in-person exam. Your exam must be complete and uploaded to Gradescope by 2:00p.

Answer the following questions and complete the exercises in RMarkdown. Please embed all of your code and push your final work to your repository. Your code must be organized, clean, and run free from errors. Remember, you must remove the # for any included code chunks to run. Be sure to add your name to the author header above.

Your code must knit in order to be considered. If you are stuck and cannot answer a question, then comment out your code and knit the document. You may use your notes, labs, and homework to help you complete this exam. Do not use any other resources- including AI assistance or other students' work.

Don't forget to answer any questions that are asked in the prompt! Each question must be coded; it cannot be answered by a sort in a spreadsheet or a written response only.

For all plots you create, a title and clearly labeled axes must be provided.

Be sure to push your completed midterm to your repository and upload the document to Gradescope. *Incorrectly uploading the exam to Gradescope will result in a 5 point deduction.* This exam is worth 50 points.

Load the libraries

```
library(tidyverse)
library(janitor)
library(naniar)
library(ggmap)
library(lubridate)
library(shiny)
```

API Key

Since you will build a map as part of the exam, take a moment to register your API key. You may need to look up your stadia maps API key online (please do that now).

```
register_stadiamaps("824e1e2a-26b0-4511-874e-b93a502f5dce", write = FALSE)
```

About the data

This study is focused on the fasting behavior of polar bears in the Southern Beaufort (SB) and Chukchi seas (CS). The data were collected from 1983-2017 and include measurements of blood urea nitrogen (BUN) and

creatinine, which are used to calculate a ratio that indicates whether a bear is fasting or not. The original paper is included in the data folder. *It may help to quickly read the abstract before starting the exam.*

The data are from: Rode, K.D., Wilson, R.R., Douglas, D.C., Muhlenbruch, V., Atwood, T.C., Regehr, E.V., Richardson, E.S., Pilfold, N.W., Derocher, A.E., Durner, G.M., Stirling, I., Amstrup, S.C., St. Martin, M., Pagano, A.M., and others, 2017, Spring fasting behavior in a marine apex predator provides an index of ecosystem productivity: *Global Change Biology*, v. 24, no. 1, p. 410–423, <https://doi.org/10.1111/gcb.13933>.

Data resources

- <https://data.usgs.gov/datacatalog/data/USGS:ASC141>
- <https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2>

The variables include:

- **bear_id**: A unique number given to each bear.
- **capture_date**: The data when the bear was tranquilized and assessed.
- **popn**: The population to which the bear belongs.
- **sex**: Sex of each bear.
- **age_class**: The age class of each bear (A=adult, S=subadult).
- **mating**: Whether or not the bear is pursuing a mate.
- **denning**: Whether or not the bear denned last year.
- **lat**: Latitude where the bear was located.
- **long**: Longitude where the bear was located.
- **urea_nitrogen**: Concentration of urea nitrogen.
- **creatinine**: Concentration of creatinine.

Load the data

Run the code chunk below to load the data, clean the names, and do some tidying.

```
polar_bears <-  
  read_csv("data/polarBear_serumUreaCreatinine_beaufortChukchi_rode.csv") %>%  
  clean_names() %>%  
  select(-den_method, -no_cubs, -cub_age) %>%  
  filter(age_class==c("A", "S")) %>%  
  mutate(sex=toupper(sex),  
         age_class=toupper(age_class))
```

1. (2 points) List the variable names and use a summary function to get an idea of the data structure.

```
names(polar_bears)
```

```
## [1] "bear_id"      "capture_date" "popn"         "sex"
## [5] "age_class"    "mating"       "denning"      "lat"
## [9] "long"         "urea_nitrogen" "creatinine"
```

```
#polar_bears
```

```
summary(polar_bears)
```

```
##      bear_id      capture_date      popn      sex
## Min.   : 697   Min.   :1983-04-23   Length:519   Length:519
## 1st Qu.:20167   1st Qu.:2000-04-10   Class :character   Class :character
## Median :20705   Median :2008-04-12   Mode  :character   Mode  :character
## Mean   :18040   Mean   :2004-07-02
## 3rd Qu.:21220   3rd Qu.:2011-03-31
## Max.   :99990   Max.   :2016-04-22
##      age_class      mating      denning      lat
## Length:519      Length:519      Length:519      Min.   :62.46
## Class :character   Class :character   Class :character   1st Qu.:68.11
## Mode  :character   Mode  :character   Mode  :character   Median :70.53
##                                     Mean   :69.86
##                                     3rd Qu.:71.00
##                                     Max.   :72.70
##      long      urea_nitrogen      creatinine
## Min.   : -172.2   Min.   : 1.00   Min.   :0.350
## 1st Qu.: -165.2   1st Qu.: 6.00   1st Qu.:0.850
## Median : -150.7   Median :11.00   Median :1.050
## Mean   : -153.7   Mean   :15.13   Mean   :1.079
## 3rd Qu.: -146.5   3rd Qu.:21.25   3rd Qu.:1.285
## Max.   : -134.8   Max.   :69.00   Max.   :2.350
```

```
#glimpse(polar_bears)
```

2. (3 points) How are NA's represented and where are they distributed in the variables? Use the `naniar` package to visualize the distribution of NA's in the data and replace any alternative representations of NA with actual NA values.

```
#NA's are represented as "unknown"
```

```
miss_var_summary(polar_bears)
```

```
## # A tibble: 11 x 3
##   variable      n_miss pct_miss
##   <chr>      <int>    <num>
## 1 bear_id          0         0
## 2 capture_date      0         0
## 3 popn              0         0
## 4 sex              0         0
## 5 age_class         0         0
```

```
## 6 mating      0      0
## 7 denning     0      0
## 8 lat         0      0
## 9 long        0      0
## 10 urea_nitrogen 0      0
## 11 creatinine 0      0
```

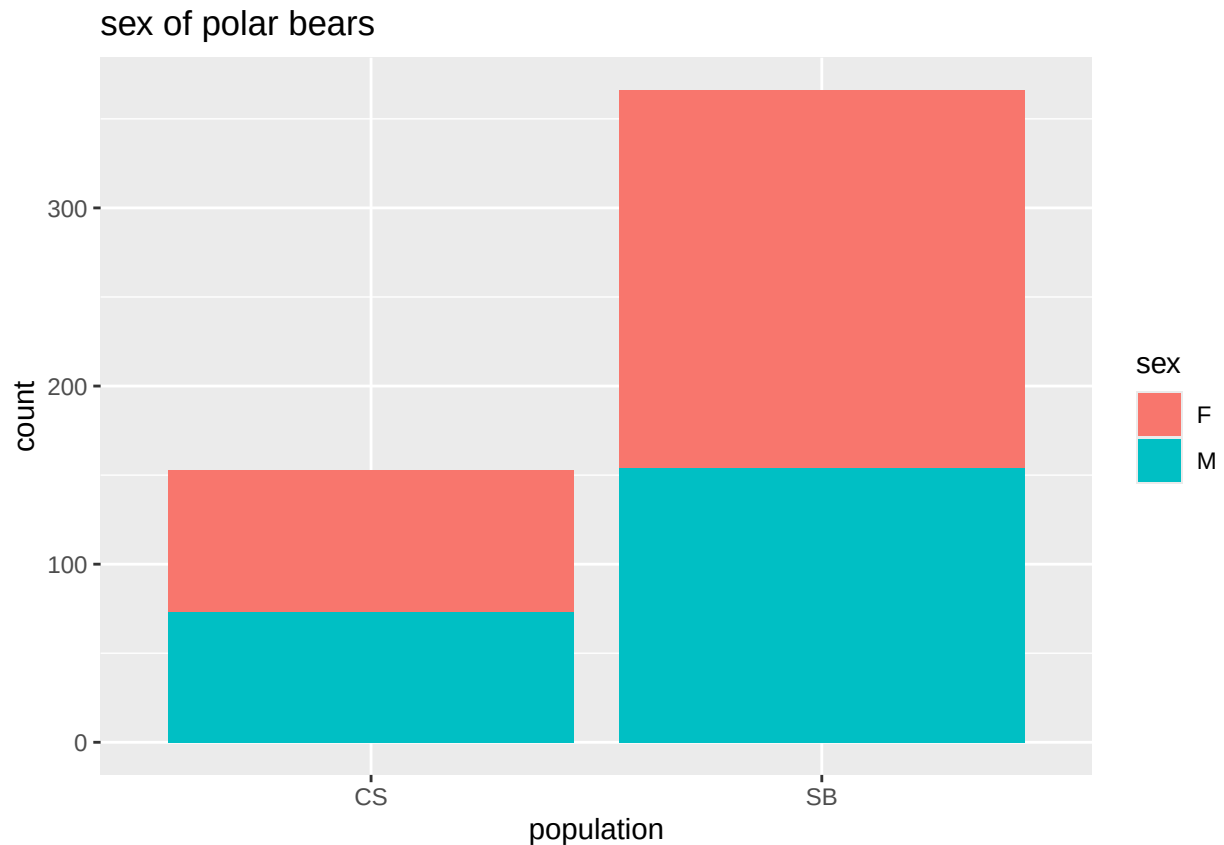
```
ppol_bears<- polar_bears %>%
  mutate(mating=na_if(mating, "unknown"))
```

```
pol_bears<- polar_bears %>%
  mutate(denning=na_if(denning, "unknown"))
```

```
polar_bears_new<- c(pol_bears, ppol_bears)
```

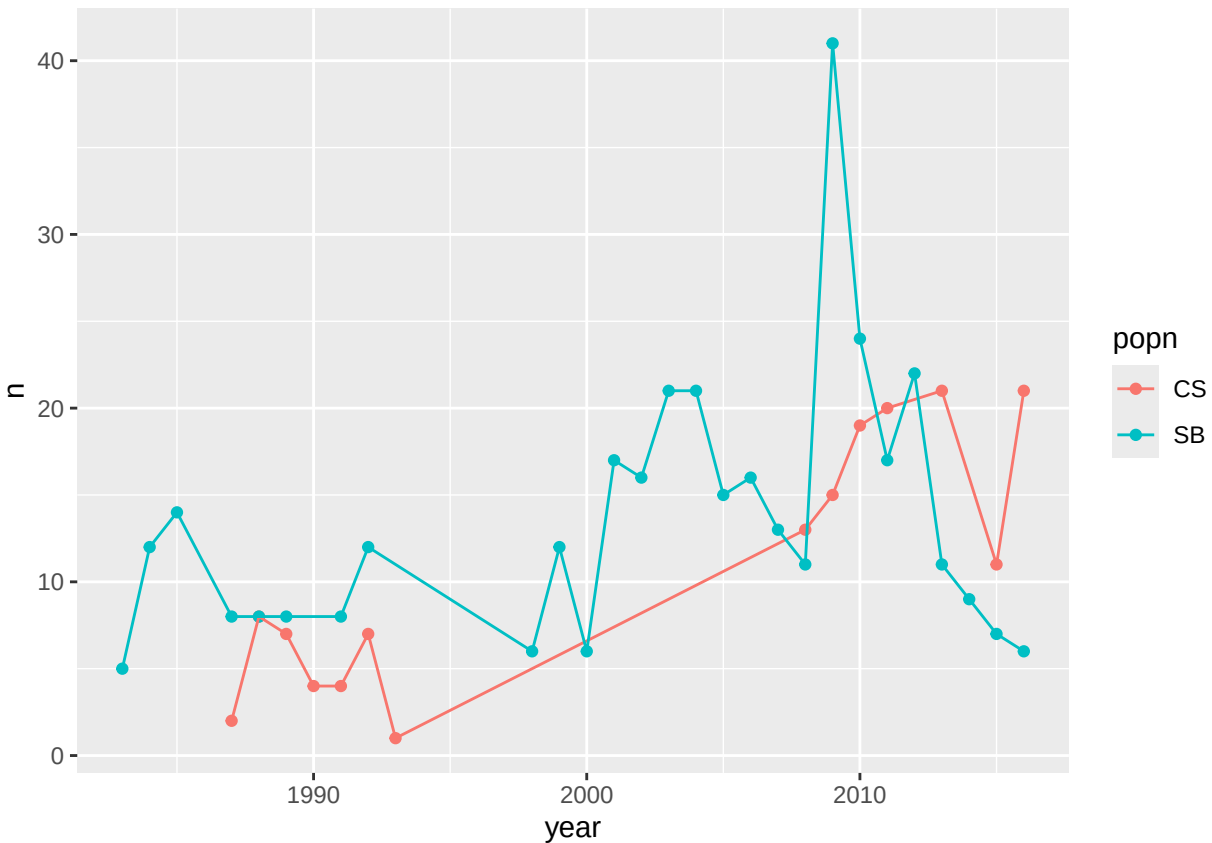
3. (6 points) Is sampling even for both sexes? Make a count the shows the number of males and females sampled by population and make a plot to visualize the counts.

```
#There are more females than males
polar_bears %>%
  group_by(sex, popn)%>%
  summarise(count=n(), .groups = "drop_last")%>%
  ggplot(aes(x = popn, y = count, fill = sex, position = "dodge"))+
  geom_col()+
  labs( title = "sex of polar bears",
        x = "population",
        y = "count")
```



4. (6 points) How many bears were sampled in each year for both populations? Make a line plot(s) to visualize sampling by year for both populations. (hint: you will need to pull apart the date to get the year in its own column)

```
#split date column then group by year
polar_bears%>%
  mutate(date = ymd(capture_date))%>%
  separate(capture_date, into = c("year", "month", "day"), sep = "-")%>%
  mutate(year = as.numeric(year)) %>%
  count(year, popn)%>%
  ggplot(aes(x= year, y= n, color = popn))+
  geom_point()+
  geom_line()
```



```
polar_bears<- polar_bears%>%
  mutate(date = ymd(capture_date))%>%
  separate(capture_date, into = c("year", "month", "day"), sep = "-")%>%
  mutate(year = as.numeric(year))
```

5. (2 points) What are the min, median, mean, and max values for urea_nitrogen and creatinine?

```
polar_bears %>%
  group_by(urea_nitrogen) %>%
  summary(mean_urea=mean(urea_nitrogen, na.rm=T),
          median_urea=median(urea_nitrogen, na.rm=T),
          min_urea=sd(urea_nitrogen, na.rm=T),
          max_urea=max(urea_nitrogen,na.rm = T),
          n=n())
```

```
##      bear_id      year      month      day
## Min.   : 697   Min.   :1983   Length:519   Length:519
## 1st Qu.:20167 1st Qu.:2000   Class :character   Class :character
## Median :20705 Median :2008   Mode  :character   Mode  :character
## Mean   :18040 Mean   :2004
## 3rd Qu.:21220 3rd Qu.:2011
## Max.   :99990 Max.   :2016
##      popn      sex      age_class      mating
## Length:519   Length:519   Length:519   Length:519
## Class :character   Class :character   Class :character   Class :character
```

```
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## denning lat long urea_nitrogen
## Length:519 Min. :62.46 Min. : -172.2 Min. : 1.00
## Class :character 1st Qu.:68.11 1st Qu.: -165.2 1st Qu.: 6.00
## Mode :character Median :70.53 Median : -150.7 Median :11.00
## Mean :69.86 Mean : -153.7 Mean :15.13
## 3rd Qu.:71.00 3rd Qu.: -146.5 3rd Qu.:21.25
## Max. :72.70 Max. : -134.8 Max. :69.00
## creatinine date
## Min. :0.350 Min. :1983-04-23
## 1st Qu.:0.850 1st Qu.:2000-04-10
## Median :1.050 Median :2008-04-12
## Mean :1.079 Mean :2004-07-02
## 3rd Qu.:1.285 3rd Qu.:2011-03-31
## Max. :2.350 Max. :2016-04-22
```

```
#Min= 1.00, Max=69.00, Median= 11.00
```

```
polar_bears %>%
group_by( creatinine) %>%
summarize(mean_cre=mean(creatinine, na.rm=T),
           median_cre=median(creatinine, na.rm=T),
           min_cre=min(creatinine, na.rm=T),
           max_cre=max(creatinine,na.rm = T),
           n=n())
```

```
## # A tibble: 53 x 6
## creatinine mean_cre median_cre min_cre max_cre n
## <dbl> <dbl> <dbl> <dbl> <dbl> <int>
## 1 0.35 0.35 0.35 0.35 0.35 1
## 2 0.4 0.4 0.4 0.4 0.4 5
## 3 0.45 0.45 0.45 0.45 0.45 3
## 4 0.47 0.47 0.47 0.47 0.47 2
## 5 0.5 0.5 0.5 0.5 0.5 5
## 6 0.55 0.55 0.55 0.55 0.55 8
## 7 0.57 0.57 0.57 0.57 0.57 3
## 8 0.6 0.6 0.6 0.6 0.6 6
## 9 0.63 0.63 0.63 0.63 0.63 1
## 10 0.65 0.65 0.65 0.65 0.65 13
## # i 43 more rows
```

```
#Min= 0.350, Max= 2.350, Median= 1.050
```

6. (3 points) As explained above, the ratio of `urea_nitrogen` to `creatinine` is used as an index of fasting. To calculate the ratio, we need to multiply `urea_nitrogen` by 0.466, then divide by `creatinine`. Create a new variable called `un_creat_ratio` that contains this calculation.

```
#calculate the creat ratio for each entry, not the mean
#summary is not needed
```

```
un_creat_ratio<- polar_bears%>%
  mutate(un_creat_ratio = (urea_nitrogen * 0.466)/creatinine)

head(un_creat_ratio)
```

```
## # A tibble: 6 x 15
##   bear_id year month day   popn sex age_class mating denning lat long
##   <dbl> <dbl> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <dbl> <dbl>
## 1    6398 1985  04    28    SB  F      A      no    yes    70.4 -148.
## 2    6384 1991  04    13    SB  F      A      no    yes    70.0 -150.
## 3    6520 1992  04    18    SB  F      A      no    yes    70.9 -145.
## 4    6559 2000  04    12    SB  F      A      no    yes    70.0 -142.
## 5   20629 2003  04    16    SB  F      A      no    yes    70.5 -145.
## 6   20876 2007  04     1    SB  F      A      no    yes    71.1 -153.
## # i 4 more variables: urea_nitrogen <dbl>, creatinine <dbl>, date <date>,
## #   un_creat_ratio <dbl>
```

7. (4 points) The authors of the study categorized bears as fasting if their `un_creat_ratio` was less than or equal to 10. Use `case_when()` to create a new variable called `fasting` that categorizes bears as fasting or not.

```
#create a new column and name it fasting
#also make a case where a bear is not fasting
un_creat_ratio<- un_creat_ratio %>%
  mutate(fasting = case_when(un_creat_ratio <= 10 ~ "fasting", un_creat_ratio > 10 ~ "not fasting")) %>%
  mutate(as.factor(fasting))

#head(fasting)
```

```
#Unload the package
detach("package:lubridate", unload = TRUE)
```

```
## Warning: 'lubridate' namespace cannot be unloaded:
## namespace 'lubridate' is imported by 'janitor' so cannot be unloaded
```

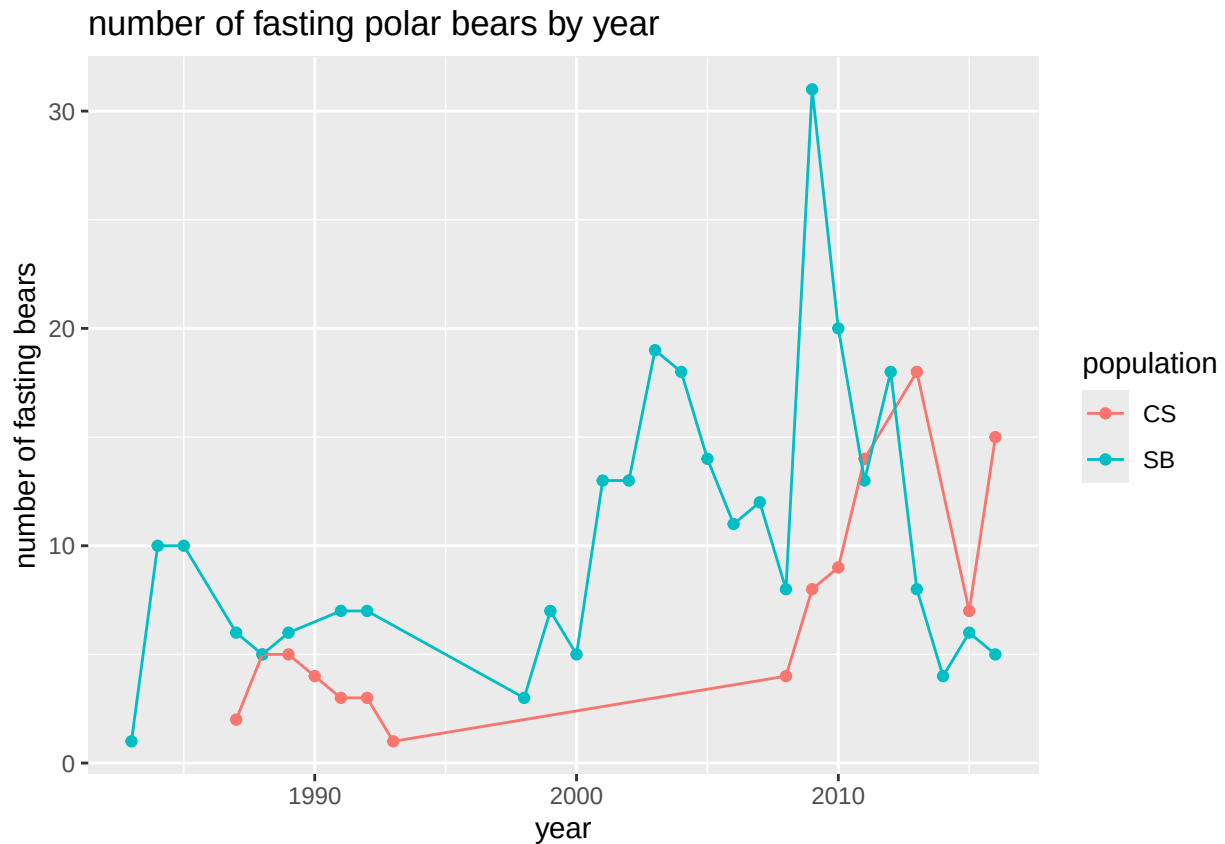
```
#simpleWarning: 'lubridate' namespace cannot be unloaded:
# namespace 'lubridate' is imported by 'janitor' so cannot be unloaded
```

8. (6 points) The authors found that the number of fasting bears increased in the SB population but decreased in the CS population. Make a plot(s) that shows the number of fasting bears by year for both populations to visualize this trend.

```
un_creat_ratio%>%
  filter(fasting == "fasting")%>%
  group_by(year, popn)%>%
  summarise(n = n())%>%
  ggplot(aes(x = year, y = n, color = popn, group = popn))+
  geom_point()+
  geom_line()+
  labs(title = "number of fasting polar bears by year",
       x = "year",
       y = "number of fasting bears",
       color = "population")
```



```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by year and popn.
## i Output is grouped by year.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(year, popn))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```



9. (8 points) Where are the polar bears used in this study located? Make a static (not leaflet) distribution map that shows the locations of bears by population. (hint: use `color=popn` in the `aes()` of `geom_point()`)

```
polar_bears %>%
  select(lat, long) %>%
  summary()
```

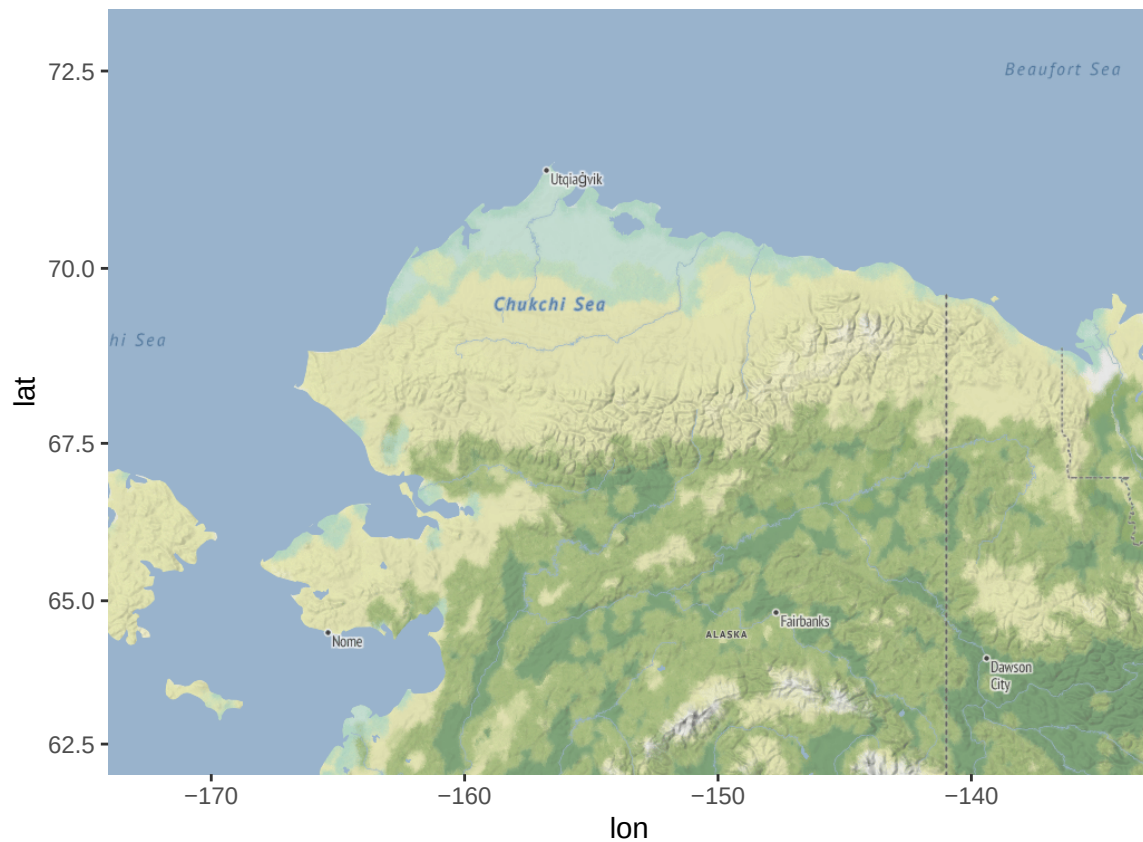
```
##      lat      long
## Min.   :62.46 Min.   : -172.2
## 1st Qu.:68.11 1st Qu.: -165.2
## Median :70.53 Median : -150.7
## Mean   :69.86 Mean   : -153.7
## 3rd Qu.:71.00 3rd Qu.: -146.5
## Max.   :72.70 Max.   : -134.8
```

```
lat <- c(62.46, 72.70)
long <- c(-172.2, -134.8)
bbox <- make_bbox(long, lat, f=0.05)
```

```
map1<- get_stadiamap(bbox, maptype = "stamen_terrain", zoom = 5)
```

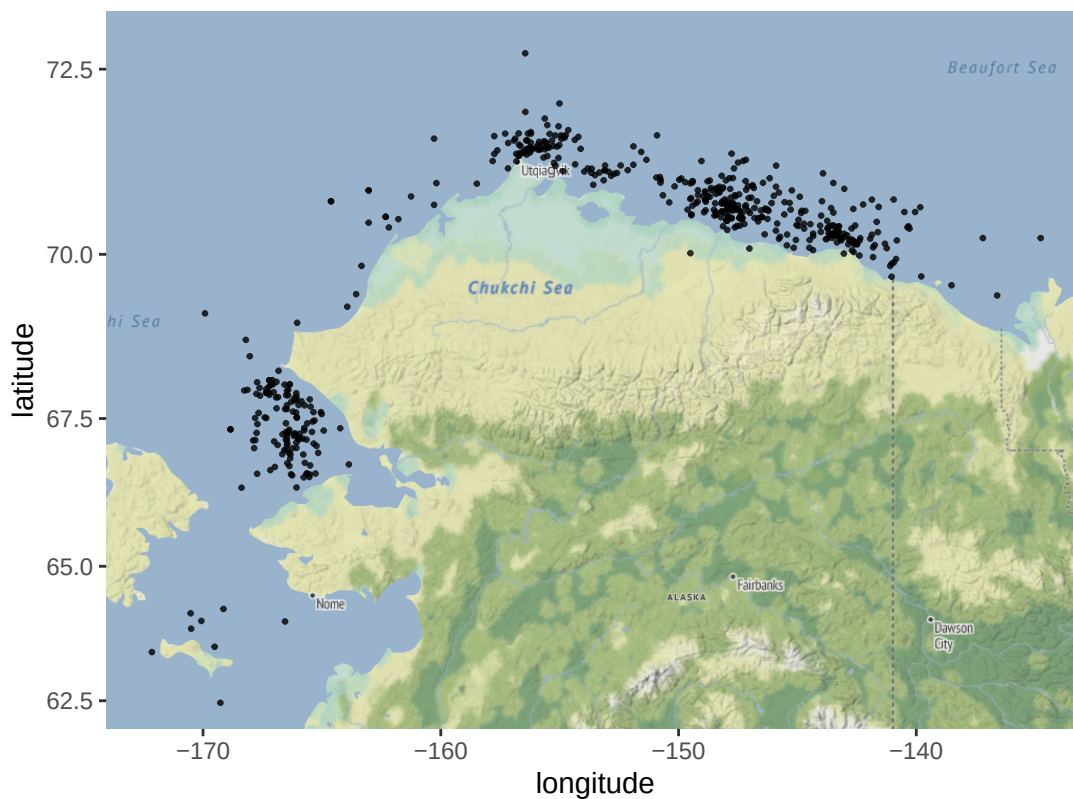
```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

```
ggmap(map1)
```



```
ggmap(map1)+
  geom_point(data = polar_bears, aes(long, lat), size = 0.5, color = "black", alpha= 0.8)+
  labs(x= "longitude",
       y= "latitude",
       title = "polar bears")
```

polar bears



10. (10 points) Build a simple shiny app that shows the range of `un_creat_ratio` values for both populations. The user should be able to select the fill variable and choose between sex, mating, age_class, denning, and fasting. Please use the code chunk below so that your exam knits.

```
library(shiny)

ui <- fluidPage(
  selectInput("x",
    "select x variable",
    choices = c("sex",
                "mating",
                "denning",
                "age_class",
                "fasting"),
    selected = "fasting"
  )

  plotOutput("plot",
    width = "500px",
    height = "400px")
)

server <- function(input, output) {
```

```

output$plot <- renderPlot({

  ggplot(data = un_creat_ratio,
        aes(x=popn,
            y=un_creat_ratio,
            fill = .data[[input$fill_var]]
        ))+
  geom_boxplot(alpha = 0.7, na.rm = T) +
  labs( title = " UN Creatine Ratio by Population",
        x = "population",
        y = "un creatine ratio")
})

}

shinyApp(ui = ui, server = server)

```

Submit the Midterm

1. Save your work and knit the .rmd file.
2. Open the .html file and “print” it to a .pdf file in Google Chrome (not Safari).
3. Go to the class Canvas page and open Gradescope.
4. Submit your .pdf file to the midterm assignment- be sure to assign the pages to the correct questions.
5. Commit and push your work to your repository.